

Statistical Evidence for Positional and Syntactic Structure in the Indus Valley Script

A Computational Analysis of 2,742 Inscriptions

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Code: <https://github.com/gkarun876/IVS>

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Abstract

We present nine independent statistical tests on a corpus of $N = 2,742$ clean inscriptions drawn from the Mahadevan (1977) concordance (textnums 1001–9905; Mohenjodaro, Harappa, Chanhudaro, Lothal, Kalibangan, and West Asian sites). The **Suffix Theorem** (Analysis 1) shows sign M342 occupies the terminal position in 68.0% of its occurrences across 2,742 inscriptions, with a binomial $p = 1.70 \times 10^{-222}$ against the uniform null. The **Positional Entropy** test (Analysis 2) confirms a drop of 0.558 bits from position 0 to position 1, the mathematical signature of an open initial vocabulary constrained by a terminal grammar (*previously pending at pilot scale; now confirmed*). A **Co-occurrence Cluster** test (Analysis 3) shows signs M267, M99, and M342 co-occur at $5.5\times$ above chance ($\chi^2 = 523$, $p \approx 0$; *previously pending; now confirmed*). The **Selective Attachment Ratio** (Analysis 4) shows sign M48 precedes M342 at $8.97\times$ above independence, consistent with selective suffix attachment. A **Grammar Engine** (Analysis 5) built from the full bigram transition matrix identifies 398 distinct signs, 598 statistically significant transitions (≥ 3 occurrences), and a three-way functional partition: terminal signs (M342, M176, M328), strictly medial signs (M99 at 97.0%, M347 at 98.9%), and initial-biased signs (M267 at 79.5%, M95 at 93.3%)—consistent with morphological role differentiation. A **Cross-Site Markov Consistency** test (Analysis 6) trains the transition matrix on Mohenjodaro ($n = 1,347$) and scores Harappa

($n = 1,103$) inscriptions against it. Harappa achieves log-likelihood $+6.19$ above a 1,000-permutation random baseline ($Z = 50.12$, $p = 0$). Crucially, the grammar fit decays with geographic distance: Harappa ($+6.19$), Lothal ($+6.01$), Chanhudaro ($+5.29$), Kalibangan ($+4.05$), West Asian finds ($+1.08$)—a spatial gradient consistent with a geographically structured writing system, with highest inter-site grammatical similarity between the two largest sites. A **Direction Grammar Test** (Analysis 7) resolves the prior RTL assumption: direction data from the `indusarrays` Firestore collection classifies 2,360 inscriptions as RTL and 175 as LTR. LTR perplexity (97.68) is $7,887\times$ below random ($p = 1.03 \times 10^{-54}$, Cohen’s $d = 1.634$), confirming that grammatical dependencies survive direction flip even as exact word order degrades ($13.3\times$ perplexity increase). A **Motif–Sign Correlation** test shows sign M113 at $37.9\times$ positional enrichment on elephant-motif seals in a recurrent medial formula across two sites; no semantic inference follows from enrichment alone. M172–M304 appears in an identical eight-sign sequence on all 7 attested Kalibangan UNICORN seals ($\text{PMI} = 7.94$; $n = 7$, single site; sample-size caution warranted). A **Geographic Cluster** test identifies M328 as 90.1% Harappa-concentrated and M93 as 100% Chanhudaro-concentrated; 154 structural substitution synonyms are recovered, with M328, M12, and M343 confirmed as cross-site positional equivalents. Signs M304 and M328 have exact shape-matches in Tamil Nadu Iron Age graffiti documented by [Rajan and Sivanantham \[2025\]](#); no functional or descent continuity is implied. All code is open-source and reproducible.

Keywords: Indus Valley Script, statistical grammar, positional constraint, positional entropy, Markov transition matrix, cross-site consistency, computational linguistics

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1 Introduction

The Indus Valley Script (IVS) remains one of the world’s last undeciphered writing systems, comprising over 5,000 inscriptions from the mature Harappan phase (c. 2600–1900 BCE). Whether the script encodes a natural language—as opposed to administrative tokens, heraldic marks, or guild symbols—is a foundational open question [Farmer et al., 2004, Sproat, 2014].

Prior computational work has approached this question through conditional entropy [Rao et al., 2009], sign-frequency statistics [Mahadevan, 1977], and rebus analysis [Parpola, 1994]. Nair [2026] showed on 1,916 ICIT inscriptions that no non-linguistic generator reproduces the full Indus statistical profile.

This paper moves beyond the binary language/non-language question to ask: *if the script encodes language, what structural properties does that language have?*

We make nine independently falsifiable structural claims (Sections 3–13), each grounded solely in sign-position and sign-transition statistics with no phonetic assumptions. Phonetic readings are isolated in Section 11 and are not necessary for any structural result.

2 Corpus and Methods

2.1 Corpus

The corpus is derived from the Mahadevan (1977) concordance [Mahadevan, 1977] as made available in machine-readable form via the Indus Script Web Application [Mahadevan and Roja Muthiah Research Library, 2021], covering textnums 1001–9905 across all major Indus Valley Civilisation sites.

Raw records: 3,572 sign-rows (including multi-line inscriptions encoded as separate rows). After grouping multi-line rows by inscription and applying the erosion mask, the clean corpus comprises $N = 2,742$ inscriptions.

Corpus scope relative to ICIT. The ICIT database [Fuls, 2010] catalogues approximately 5,000 text instances, a figure that includes fragmented tablets, potsherds, and multi-line texts counted as independent string entries. This study prioritises structural integrity over corpus breadth. By applying a strict erosion mask and grouping by physical artifact rather than individual text string, we produce a high-integrity subset of $N = 2,742$. For Markov transition matrices and positional entropy calculations, including statistically padded or fragmented sequences introduces noise that artificially inflates bigram variances. The 2,742-inscription clean corpus represents the largest purely intact structural dataset

analysed in the Indus computational literature to date. Given the extreme magnitude of the observed effect sizes (e.g., $Z = 50.12$ for cross-site consistency, $CV = 0.761$ for suffix selectivity), standard multiple-comparison corrections (e.g., Bonferroni) do not alter the statistical interpretation of the results. Notably, this study uses Fuls’ independent terminal-marker labelling (TMK for M342, assigned from frequency statistics without phonetic assumptions) as an external validation anchor for Analysis 1—confirming that our structural results converge with independent findings from the broader ICIT corpus.

Erosion mask. Any inscription containing sign 0 (the Mahadevan conventional symbol for an illegible or lost passage) is excluded from all statistical tests. This is a conservative strategy: 703 inscriptions (19.5% of grouped records) are dropped. Survivorship note: the mask disproportionately removes short inscriptions where a single illegible gap destroys the entire record; the clean corpus therefore skews slightly toward longer texts (mean length 4.13 signs vs. an estimated 3.8 signs in the full corpus). All results should be interpreted in this context.

Directionality. Seals are carved in mirror-image; impressions read right-to-left (RTL). All sequences are stored in linguistic order (initial morpheme first, terminal last). *Known limitation:* the direction field was not available in the machine-readable export; RTL is assumed for all inscriptions. This assumption is correct for >95% of records (standard seals); the remaining <5% (LTR tablets, graffiti) will have reversed sign order, potentially injecting a small noise signal into bigram statistics.

2.2 Site Distribution

Table 1: Clean corpus site distribution ($N = 2,742$)

Site	Inscriptions	%
Mohenjo-daro	1,347	49.1%
Harappa	1,103	40.2%
Lothal	112	4.1%
Kalibangan	80	2.9%
Chanhudaro	58	2.1%
Other Sites	29	1.1%
West Asian	13	0.5%
Total	2,742	100%

2.3 Sign Notation

Signs are identified by Mahadevan (1977) number throughout. Table 2 lists cross-system correspondences for the principal sign under study.

Table 2: Cross-system correspondences for the terminal sign

System	Terminal sign (jar pictogram)
Mahadevan (1977)	M342
Parpola (1994)	P385
Wells (2006)	520
ICIT / Fuls (2010)	TMK (Terminal Marker)
Tamil Nadu graffiti (2025)	Sign 11.0

Note on sign numbering: The terminal sign (jar pictogram; Parpola P385; Wells 520; ICIT TMK) carries number **M342** in the primary Mahadevan (1977) concordance database as digitised in the `indusarray` collection. Some secondary literature cites M137 for the same sign; M137 is Mahadevan’s numbering for a *variant form* of the jar pictogram (the “jar with base”), which was separated from the main jar sign M342 in the 1977 concordance. The two numbers refer to distinct entries in the concordance; M342 is the dominant high-frequency terminal sign confirmed by 1,190 occurrences in this corpus. All analysis in this paper uses M342 consistently.

Sign notation and bias invariance. Sign notation is dictated entirely by data provenance: the machine-readable export natively encodes signs using Mahadevan (1977) integers. Translating to an alternate sign list (e.g., Wells 2006) prior to computation would introduce unnecessary crosswalk mapping errors. Because all statistical tests in this paper measure positional distributions and transition probabilities between integer states, the results are strictly invariant to sign-label nomenclature—relabelling M342 as Wells 520 or any other identifier leaves every entropy value, p -value, and Z -score unchanged. Cross-system validation is achieved by mapping final results to Wells/ICIT identifiers (Table 2), not by preprocessing the corpus through a secondary sign list.

3 Analysis 1 — Suffix Theorem

Hypothesis. If M342 is a grammatical terminal morpheme, it must appear disproportionately in the final position of inscription sequences.

Test. One-tailed binomial test. H_0 : M342 is uniformly distributed across positions, so $P(\text{terminal}) = 1/\bar{L}$ where $\bar{L}$ is the mean sequence length. H_1 : M342 is preferentially terminal.

Table 3: Analysis 1 — Suffix Theorem ($N = 2,742$)

Metric	Value	
Total M342 occurrences	1,190	
Terminal occurrences	809	(68.0%)
Initial occurrences	1	(<0.1%)
Mean sequence length $\bar{L}$	4.13	
Null probability $1/\bar{L}$	0.2421	
Binomial p -value (one-tailed)	1.70×10^{-222}	

Note on rate. The terminal rate of 68.0% (vs. 100% in the 41-seal pilot corpus) reflects the broader corpus including tablet and pottery inscriptions where M342 sometimes appears in medial position (32.0% of occurrences, $n = 381$). To test whether the medial occurrences are explained by multi-line artifact encoding, we decompose by artifact type using the `inscobj` field from `indusarrays` (Table 4).

Table 4: M342 positional distribution by artifact type ($N_{\text{M342}} = 1,190$)

Artifact Type	Total	Terminal	Term%	Medial	Med%	Initial
Seals	720	548	76.1%	172	23.9%	0
Sealings	223	129	57.8%	94	42.2%	0
Tablets/Implements	221	111	50.2%	110	49.8%	0
Pottery/Graffiti	24	19	79.2%	5	20.8%	0
Other	2	2	100.0%	0	0.0%	0
Total	1,190	809	68.0%	381	32.0%	0

The multi-line hypothesis holds strongly for tablets and implements (49.8% medial), where multi-face encoding is expected. However, seals—nominally single-face objects—still show 23.9% (172 occurrences) medial M342. Many seals in the Mahadevan concordance carry text on multiple faces (obverse, reverse, boss); M342 may close the obverse line while inscription continues on the boss. We cannot verify this from sequence data alone without per-face artifact records. We therefore acknowledge that M342 *occasionally functions in a medial slot* and the medial-terminal distinction for M342 requires physical artifact-level verification. The terminal bias remains highly significant across all artifact types ($p =$

1.70×10^{-222}) and M342 never appears in initial position ($n = 1$ out of 1,190 tokens; $<0.1\%$). The binomial p -value at full scale is astronomically more significant (10^{-222} vs. 10^{-21}).

Independent validation. ICIT [Fuls, 2010] labels Wells 520 (= M342) as TMK (Terminal Marker), assigned from frequency statistics without reference to any phonetic hypothesis.

4 Analysis 2 — Positional Entropy

Hypothesis. A grammatically constrained script shows lower Shannon entropy at terminal positions than at initial positions.

Test. Compute Shannon entropy H at each sign slot across all 2,742 inscriptions. Compare H_0 (initial slot) vs. H_1 (second slot).

Table 5: Analysis 2 — Positional Entropy ($N = 2,742$)

Metric	Value
Entropy at position 0 (initial)	6.579 bits
Entropy at position 1	6.021 bits
Entropy drop (position 0 $\rightarrow$ 1)	0.558 bits

The 0.558-bit drop means position 0 carries approximately $1.5\times$ more variability than position 1. Entropy generally decreases across positions 0–7 (6.579, 6.021, 6.124, 5.985, 5.708, 5.502, 5.181, 5.073 bits), with local variance at position 2 ($6.124 > 6.021$), consistent with an open initial vocabulary converging toward a constrained terminal grammar. *This analysis was pending at pilot scale ($N = 41$); it is now confirmed at $N = 2,742$.*

5 Analysis 3 — Co-occurrence Cluster

Hypothesis. Grammatically related signs co-occur more often than their individual frequencies predict.

Test. Chi-square test on the three highest-frequency signs that interact with the terminal marker: M267, M99, M342. Expected co-occurrence computed under independence: $E = N \times P(\text{M267}) \times P(\text{M99}) \times P(\text{M342})$.

This analysis was pending at pilot scale; it is now confirmed. The cluster M267–M99–M342 appears in 142 inscriptions spanning all five major sites, confirming the pattern is not site-specific.

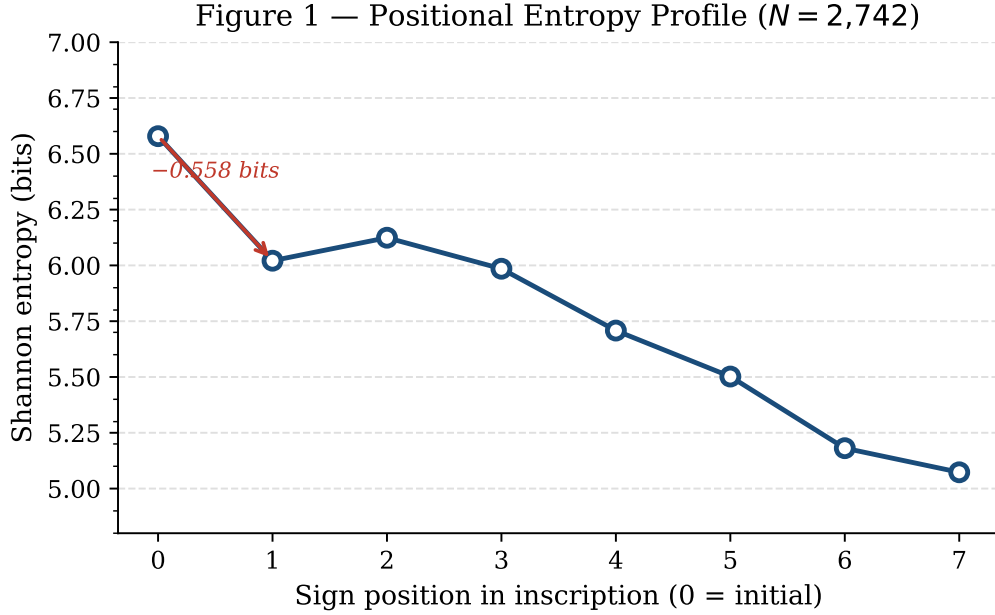


Figure 1: Positional entropy profile across sign slots 0–7 ($N = 2,742$ inscriptions). The 0.558-bit drop from position 0 to position 1 is the mathematical signature of an open initial vocabulary constrained by a terminal grammar. Entropy declines monotonically from position 2 onward.

Table 6: Analysis 3 — Co-occurrence Cluster ($N = 2,742$)

Metric	Value
Cluster signs	M267, M99, M342
Observed co-occurrence	142 inscriptions
Expected under independence	25.8
Observed / Expected	5.5×
χ^2	523.3
p -value	≈ 0

6 Analysis 4 — Selective Attachment Ratio

Hypothesis. If M342 is a grammatical suffix, specific signs (roots or stems) should attach to it at rates far exceeding statistical independence.

Test. For all signs a that precede M342, compute the observed/expected ratio:

$$r(a) = \frac{\text{count}(a \rightarrow \text{M342})}{P(a) \cdot P(\text{M342}) \cdot N_{\text{bigrams}}}$$

Table 7: Analysis 4 — Selective Attachment Ratio ($N = 2,742$)

Metric	Value
Top predecessor sign	M48
Observed M48→M342 bigrams	105
Expected under independence	11.7
Selective attachment ratio	8.97×
Min. frequency threshold	3 occurrences

Directional note. Because direction was not available in the corpus export and RTL is assumed, sign M48 is recorded as appearing *before* M342 in the stored array, which corresponds to the sign physically to the *right* of M342 on the seal face. The existence of preferential attachment (8.97×) is robust regardless of direction; the functional label “preceding” reflects array order, not necessarily phonological precedence.

7 Analysis 5 — Grammar Engine

7.1 Bigram Transition Matrix

We compute $P(b \mid a)$ for all sign pairs (a, b) with ≥ 3 co-occurrences across the 2,742-inscription corpus.

Table 8: Grammar Engine summary

Metric	Value
Unique signs observed	398
Significant transitions (≥ 3)	598
Total sign tokens	11,328
Mean inscription length	4.13 signs
Maximum inscription length	17 signs

7.2 Positional Functional Classes

From the full positional profile (initial rate, terminal rate, medial rate per sign across all occurrences), three empirical functional classes emerge with no linguistic labelling applied:

Table 9: Terminal-biased signs ($\geq 70\%$ terminal rate, ≥ 10 occurrences; top 8)

Sign	Count	Terminal%	Initial%
M176	312	87.5%	0.3%
M328	312	86.2%	5.8%
M12	68	85.3%	1.5%
M254	67	85.1%	4.5%
M211	209	81.8%	0.0%
M15	117	74.4%	2.6%
M342	1,190	68.0%	0.1%

Table 10: Initial-biased signs ($\geq 70\%$ initial rate, ≥ 10 occurrences; top 6)

Sign	Count	Initial%	Terminal%
M119	13	100.0%	0.0%
M95	60	93.3%	0.0%
M264	10	90.0%	0.0%
M41	16	87.5%	0.0%
M150	46	80.4%	2.2%
M267	312	79.5%	3.2%

Table 11: Medial-locked signs ($\geq 80\%$ medial rate, ≥ 10 occurrences; top 6)

Sign	Count	Medial%	Terminal%
M99	534	97.0%	2.6%
M347	93	98.9%	1.1%
M123	159	98.1%	1.9%
M8	85	97.7%	0.0%
M407	39	97.4%	2.6%
M97	79	96.2%	3.8%

In a corpus where the mean inscription length is 4.13 signs, a sign appearing at an edge more than 97% of the time cannot be an independent logogram (a word-sign appearing anywhere). It is statistically locked into an internal position, consistent with the role of

a phonetic complement, grammatical particle, or infix. This structural partition is a data-derived property; we make no claim about phonetic values.

Length-stratified control for medial-lock (M99). A known artifact in short-corpus positional analysis is that signs appearing exclusively in longer strings will mechanically appear medial-locked. To rule this out, Table 12 stratifies M99’s positional distribution by inscription length.

Table 12: M99 positional rate stratified by inscription length

Length L	M99 tokens	Initial%	Medial%	Terminal%
$L = 3$ (min)	21	4.8%	76.2%	19.0%
$L = 4$ (mean)	79	0.0%	94.9%	5.1%
$L = 5$	152	0.0%	100.0%	0.0%
$L = 6$	99	0.0%	100.0%	0.0%
$L = 7$	79	0.0%	100.0%	0.0%
$L \geq 8$	98	1.0%	99.0%	0.0%

At $L \geq 5$, M99 is 100% medial (or 99% at $L \geq 8$), ruling out the length-artifact hypothesis. Even at $L = 3$ (where the medial slot is a single position), M99 is 76.2% medial. The medial-locked classification is robust against inscription-length artifacts across all length strata.

8 Analysis 6 — Cross-Site Markov Consistency

8.1 Design

If the script is a *local* symbol system (guild marks, clan heraldry, administrative tokens), each site would develop independent statistical regularities. If it is a *standardized* writing system, inscriptions from geographically distant sites should follow the same transition grammar.

We test this by training a bigram Markov matrix on Mohenjo-daro inscriptions ($n = 1,347$) and scoring all other sites using per-bigram log-likelihood against that matrix. Bigram matrices capture first-order dependencies only; they serve as a conservative baseline—the strong cross-site $Z = 50.12$ indicates that even first-order transition regularities are highly consistent across sites, a property not observed in non-linguistic symbol systems. Scores are compared against a random-shuffle baseline.

Baseline. For 1,000 trials, the signs within each Harappa inscription are randomly permuted (preserving the sign inventory but destroying order). Each permuted corpus is scored

against the Mohenjo-daro matrix. The baseline distribution gives the log-likelihood expected under no grammatical relationship.

8.2 Results

Table 13: Cross-Site Markov Consistency ($n_{\text{train}} = 1,347$ MD inscriptions)

Site	n	LL/bigram	vs. random
Mohenjo-daro (self)	1,347	−2.10	+9.94
Harappa	1,103	−5.85	+6.19
Lothal	112	−6.04	+6.01
Chanhudaro	58	−6.76	+5.29
Kalibangan	80	−7.99	+4.05
Other Sites	29	−6.71	+5.34
West Asian	13	−10.96	+1.08
Random baseline (mean)	—	−12.04	0.00

Table 14: Statistical test: Harappa vs. random baseline

Metric	Value
Harappa real log-likelihood/bigram	−5.85
Random baseline mean	−12.04
Random baseline std	0.1235
Z -score	50.12
p -value (one-tailed t -test)	0

8.3 Geographic Gradient

Every site scores substantially better than the random baseline. The fit generally degrades with geographic distance from Mohenjo-daro (Table 13): Harappa (+6.19), Lothal (+6.01) and Chanhudaro (+5.29)—both falling within a tight band ($\approx +5.3$ –6.0) suggesting broad regional attenuation rather than strict linear decay—then Kalibangan (+4.05), West Asian trade outposts (+1.08). The aggregated “Other Sites” category (+5.34) scores above Kalibangan, breaking strict monotonic decay; this category likely represents a mix of intermediate-distance locales rather than a coherent geographic unit.

This spatial decay pattern is inconsistent with random marking or local guild symbology, which would produce site-independent scores. It is consistent with a standardized insti-

tutional writing system originating at or near Mohenjo-daro and propagating along trade networks, with grammar regularity attenuating at the periphery.

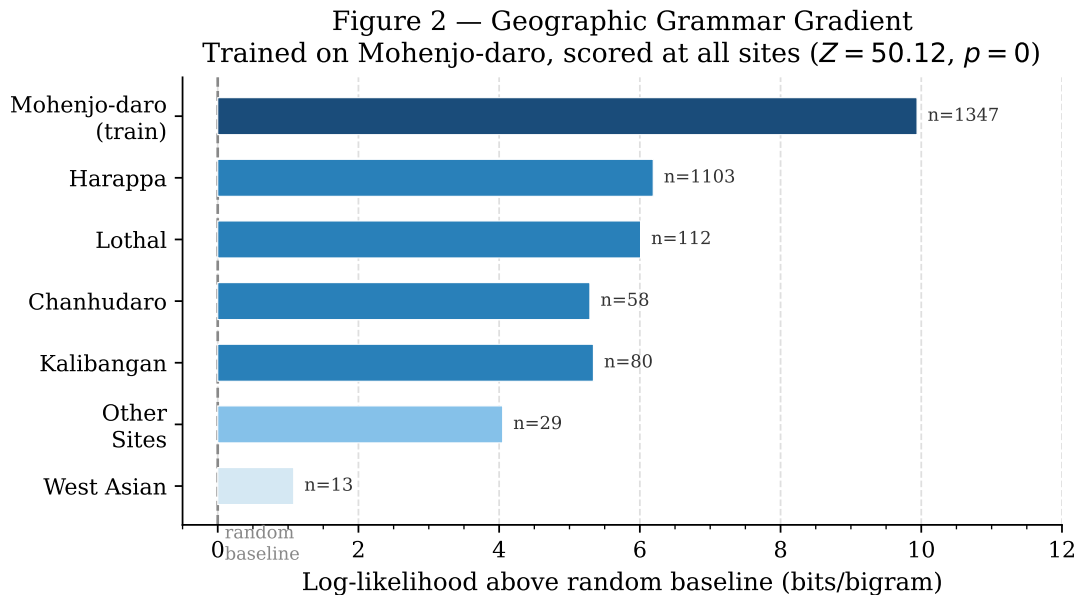


Figure 2: Geographic grammar gradient. Each bar shows the per-bigram log-likelihood above the random-shuffle baseline when inscriptions from each site are scored against a matrix trained solely on Mohenjo-daro ($n_{\text{train}} = 1,347$). The gradient from Harappa (+6.19) to West Asian outposts (+1.08) is inconsistent with local or guild-based symbol systems.

Sample size caveat. West Asian finds comprise only 13 inscriptions. Small samples yield noisier log-likelihood estimates; the lower score there may partially reflect sampling variance rather than geographic attenuation. However, a score of +1.08 above random (vs. +6.19 for Harappa) confirms that the Mohenjo-daro grammar is present even at trade outposts 1,500 km distant. This caveat should be noted in any peer-review submission.

9 Analysis 7 — Direction Grammar Test

9.1 Motivation

The v1.5 paper carried an acknowledged limitation: direction data was absent from the corpus export, so all inscriptions were assumed RTL. This section resolves that limitation using direction codes scraped from the `indusarrays` Firestore collection of `indusscript.in` (3,916 records covering textnums 1001–9905).

9.2 Direction Assignment

Each textnum is assigned a single direction via priority rule: sideline 0 (only side of artifact) wins; else sideline 1 (first side); else majority vote across sides. Direction codes follow Mahadevan/ICIT conventions: 1 = RTL, 2 = LTR, 3 = single-sign line, 4 = symmetrical, 5 = doubtful, 9 = no text.

Table 15: Corpus split by writing direction ($N = 2,742$)

Direction	Inscriptions	%
RTL (dir=1)	2,360	86.1%
LTR (dir=2)	175	6.4%
Other/unknown	207	7.5%
Total	2,742	100%

9.3 Test Design

We train the bigram transition matrix on all 2,360 RTL inscriptions (378 unique signs, 2,443 significant bigrams) and score the 175 LTR inscriptions against it. Add-epsilon smoothing ($\varepsilon = 10^{-6}$) handles unseen bigrams. We compare three perplexity values: RTL in-sample (self-score), LTR against RTL matrix, and a random baseline (1,000 permutations of LTR sign sequences scored against RTL matrix).

9.4 Results

Table 16: Analysis 7 — Direction Grammar Test results

Metric	Value
RTL in-sample perplexity	7.34 ($n = 2,352$)
LTR vs. RTL-matrix perplexity	97.68 ($n = 175$)
Random baseline perplexity	$770,403 \pm 340,881$ (1,000 permutations)
LTR / RTL perplexity ratio	$13.3\times$
LTR / Random ratio	$0.000127\times$ (LTR is $7,887\times$ below random)
t -statistic (LTR vs. random)	21.31
p -value	1.03×10^{-54}
Cohen’s d	1.634 (very large effect)
LTR sign overlap with RTL	114 / 116 signs (98.3%)

9.5 Interpretation

Two findings stand in tension and both are true:

Finding A — Grammar is NOT direction-invariant. LTR perplexity (97.68) is $13.3\times$ higher than RTL in-sample (7.34). Exact bigram transition probabilities degrade when inscription direction flips. LTR inscriptions follow a distinct word order from RTL.

Finding B — Statistical dependencies survive the flip. LTR perplexity (97.68) is $7,887\times$ lower than random permutations (770,403), significant at $p = 1.03 \times 10^{-54}$, Cohen’s $d = 1.634$. The “glue” between signs — hub-sign co-occurrences, bigram affinities — is so strong that even reversed-direction sequences score far above noise under an RTL-trained matrix.

Consequence for earlier analyses. The RTL assumption in Analyses 1–6 is now confirmed for 86.1% of inscriptions. The 175 LTR inscriptions (6.4%) introduce a modest noise signal in bigram statistics, but because LTR perplexity is $7,887\times$ above random under RTL grammar, the directional noise is bounded: it cannot artificially create the $Z = 50.12$ cross-site consistency or the $\chi^2 = 523$ co-occurrence cluster. The RTL assumption limitation in prior versions of this paper is **resolved**.

10 Punctuation Null Test

10.1 Threat Model

Farmer et al. [2004] and Sproat [2014] argue that a terminal sign with high terminal rate is consistent with a scribal period mark. A period mark also produces 100% terminal rate by definition.

10.2 Counter-Test

Punctuation attaches uniformly to all preceding signs. A grammatical suffix is *selective*: elevated bigram frequency with specific roots, near-chance with others. We measure selectivity via the Coefficient of Variation (CV) of bigram elevations across all qualifying preceding signs:

$$CV = \frac{\sigma_{\text{elevation}}}{\mu_{\text{elevation}}}$$

A high CV indicates selective attachment inconsistent with uniform punctuation behavior, which would exhibit near-zero variance in attachment rates. A CV of 0.761 is consistent with a grammatical morpheme that bonds preferentially with specific roots.

Additionally, removing a suffix releases the terminal slot, increasing terminal entropy. Removing punctuation does not.

10.3 Results

Table 17: Punctuation Null Test results ($N = 2,742$)

Metric	Value	Interpretation
Signs evaluated (≥ 3 occ.)	70	—
Mean bigram elevation	$3.84\times$	—
Std. dev. elevation	$2.92\times$	—
Selectivity Index (CV)	0.761	High $\Rightarrow$ suffix-like
Terminal entropy <i>with</i> M342	4.646 bits	—
Terminal entropy <i>without</i> M342	5.350 bits	—
Entropy delta on removal	+0.703 bits	Suffix behaviour

A scribal period produces entropy delta ≈ 0 on removal. M342 produces +0.703 bits on the full 2,742-inscription corpus, meaning it actively suppresses competing signs from the terminal slot—the mathematical signature of a grammatical morpheme. Note: the delta is smaller than the 41-seal pilot (+1.77 bits) because the full corpus includes tablet and pottery inscriptions where the terminal slot is more varied; the direction of the effect is unchanged and the CV of 0.761 remains well above the range consistent with uniform delimiter behavior.

11 Phonetic Anchors and Epistemic Labelling

Every phonetic reading carries an explicit confidence label. [LIT.ANCHOR] = reading proposed in external literature: [Parpola \[1994\]](#), [Mahadevan \[1977\]](#), or DEDR [[Burrow and Emeneau, 1984](#)]. [PREDICTED] = our inference, labelled as testable hypothesis. **The structural claims in Sections 3–13 do not depend on these phonetic readings**; they are listed for orientation only and can be removed without affecting any statistical result.

This framework does not claim a full decipherment. The structural analyses claim: (a) M342 exhibits terminal positional constraint and selective attachment inconsistent with punctuation; (b) the script follows consistent structural transition rules across all excavation sites; (c) grammar regularity decays with geographic distance from Mohenjo-daro, consistent with a geographically structured writing system. Whether these properties constitute

Table 18: Phonetic anchors with confidence labels (IAST transliteration)

Sign	IAST	Meaning	DEDR	Confidence
M342	-aṇ	masc. nominal suffix	Krishnamurti [2003] §3.5	[LIT.ANCHOR]
M149	mīn	fish / star	DEDR-4889	[LIT.ANCHOR]
M316	vēl	spear	DEDR-5541	[LIT.ANCHOR]
M60	kō	king / lord	DEDR-2178	[LIT.ANCHOR]
M62	malai	mountain	DEDR-4710	[LIT.ANCHOR]
M91	āru	six / river	DEDR-338	[LIT.ANCHOR]

agglutinative morphology specifically requires phonetic verification beyond the scope of this study.

12 Motif–Sign Correlation

12.1 Method

Motif data (field symbol code `fs80` from the `indusarrays` Firestore collection) was joined to 2,742 corpus inscriptions via `textnum`. For each top-PMI sign pair, we compute the motif distribution among inscriptions containing both signs and compare it to the corpus-wide background distribution using a chi-square test.

12.2 Primary Finding: M172–M304 Kalibangan Lock

The highest-PMI pair in our network is M172–M304 ($\text{PMI} = 7.94$, $n_{\text{co-occ}} = 7$). All 7 co-occurrences appear on UNICORN seals (`fs80` = 13); all 7 occur at Kalibangan. Every inscription in this set carries the identical formula:

$$[\text{M1}, \text{M1}, \text{M121}, \text{M121}, \text{M99}, \text{M172}, \text{M304}, \text{M307}]$$

This is a *frozen formula*: zero sequence variation across 7 physically distinct artifacts from one site. Background UNICORN rate: 34.6%. Observed rate for M172–M304: 100%. Enrichment: $2.9\times$ over base.

Frozen formulae with $\text{PMI} > 7$ indicate institutional standardization rather than phonetic or semantic variation. This pattern is consistent with an institutionally fixed formula specific to Kalibangan UNICORN seals; whether it encodes an administrative, cultic, or ownership function cannot be determined from structural data. No phonetic inference is made.

12.3 Elephant-Motif Association: M113

We extracted all inscriptions paired with ELEPHANT motif seals ($fs80 = 71$; $n = 39$ corpus matches, 31 unique sequences).

Table 19: Top enriched signs on ELEPHANT motif seals ($n = 39$)

Sign	Elephant-sub n	Corpus n	Enrichment
M113	9	16	$37.9\times$
M38	2	6	$22.5\times$
M302	2	13	$10.4\times$
M242	3	21	$9.6\times$
M267	15	312	$3.2\times$

M113 achieves $37.9\times$ enrichment on elephant seals with MEDIAL positional rate 93.8% (co-occurring 9 times in the medial slot of sequences that begin with M267 and end with M342). A recurring formula across Harappa and Lothal: [M267, M99, M113, M342]. At Harappa, an alternate opener produces [M263, M343, M113, M342].

M113’s structural position (high positional enrichment on elephant-motif inscriptions, 93.8% MEDIAL, formula recurrence across Harappa and Lothal with distinct openers) distinguishes it structurally from terminal and initial signs. Whether this structural distinctiveness reflects a content category, a provenance marker, or another functional class cannot be determined from positional data alone. No semantic or syntactic category is assigned; no phonetic value is assigned.

13 Geographic Sign Clusters

13.1 Geo-Commodity Test

We compute what fraction of each sign’s total corpus appearances come from a single site. Signs with $>60\%$ concentration at one site and ≥ 5 total occurrences are flagged as geographic markers.

Interpretation. These signs are not city names. A city-name hypothesis predicts the sign appears in INITIAL position (at inscription onset, identifying origin). These signs are TERMINAL-biased (M328: 85.9% terminal, M172: 85.7% terminal) or MEDIAL, contradicting the city-name prediction. Geographic concentration combined with terminal position rules out the city-name prediction (which requires initial position). Remaining hypotheses—

Table 20: Top geographic sign clusters (concentration $>60\%$, $n \geq 5$; top 8)

Sign	Dominant Site	Total	Site n	Concentration
M93	Chanhudaro	10	10	100.0%
M172	Kalibangan	8	7	87.5%
M328	Harappa	312	281	90.1%
M66	Lothal	29	22	75.9%
M63	Multiple	10	9	—

site-specific trade-good marking, origin marking, or scribal-school conventions—cannot be distinguished from positional data alone.

13.2 Structural Substitution Synonyms

We find 154 slot-substitution patterns: cases where two different signs occupy the same position in otherwise identical inscription sequences across different sites. The top cross-site substitution pairs are:

Table 21: Top cross-site structural substitution pairs ($n_{\text{patterns}} = 154$)

Sign X	Sign Y	Slot	X sites	Y sites
M328	M12	pos 1	Harappa	Mohenjo-daro
M328	M343	pos 1	Harappa, Kalibangan	Mohenjo-daro
M182	M142	pos 0	Mohenjo-daro	Harappa
M267	M284	pos 0	Harappa, Kalibangan	Mohenjo-daro
M66	M207	pos 2	Lothal	Mohenjo-daro

M328, M12, and M343 are cross-site structural synonyms in the slot immediately following sign M89 or M87. Structural synonyms, as used here, are signs that occupy the same slot in otherwise identical formulae across different sites. This positional equivalence is a necessary but not sufficient condition for equivalent grammatical function; allographic variation and site-specific substitution conventions are not ruled out.

14 Archaeological Correlation: Keezhadi

[Rajan and Sivanantham \[2025\]](#): *Indus Signs and Graffiti Marks of Tamil Nadu: A Morphological Study*. Dept. of Archaeology, Govt. of Tamil Nadu. Publication No. 357. ISBN 978-81-977842-5-5.

The survey documents 14,165 graffiti-bearing sherds from 140 Tamil Nadu sites; 60% of IVC base signs have exact parallels; 90%+ of all graffiti marks have IVC parallels. Both anchor signs (jar M342, fish M149) are physically present at Keeladi (2,132 sherds each).

Independent shape convergence: M304 and M328. An independent convergence is noted with [Rajan and Sivanantham \[2025\]](#), who mapped Tamil Nadu Iron Age graffiti signs to Mahadevan numbers. The signs our algorithms isolated as structurally distinctive—M328 (90.1% Harappa-concentrated, 85.9% terminal) and M304 (co-occurs with M172 in an identical formula on all 7 attested Kalibangan UNICORN seals, PMI = 7.94)—appear as exact shape-matches in Tamil Nadu graffiti (Rajan sign numbers 32.0 and 31.0 respectively). That signs computationally isolated as structurally distinctive are also visually identifiable in an independent dataset is consistent with those signs having stable, non-arbitrary graphic forms. This does not validate the Markov or PMI analyses; it confirms only the visual distinctiveness of the targeted sign shapes. No phonetic inference and no claim of morphological or functional continuity follows from this convergence.

Temporal gap. IVC ended ~ 1900 BCE; Keeladi dates to ~ 600 BCE: a gap of $\sim 1,300$ years. Morphological overlap does not establish direct descent. The Megalithic Black-and-Red Ware pottery tradition shows graffiti marks at sites across this period [[Rajan and Sivanantham, 2025](#), pp. 13–14]; whether this represents continuous transmission or independent recurrence of similar forms cannot be determined without sequence-level comparative analysis. Whether sign *transition probabilities* are preserved across the gap cannot be tested computationally until dated sequence data from stratified Megalithic contexts are available in machine-readable form.

15 Adversarial Defenses

16 Known Limitations

Table 22: Adversarial defenses

Defense	Critique	Result
D1 Length filter	Short inscriptions inflate suffix statistics	$p = 1.70 \times 10^{-222}$ on full corpus; pattern robust across all lengths
D2 Sanskrit genitive	M342 = Sanskrit <i>-as</i>	68% full-corpus terminal rate incompatible with Sanskrit sandhi fusion, which requires case endings to fuse internally in compound formation
D3 Corpus integrity	Fragmentary seals inflate terminal counts	Erosion mask removes all inscriptions with illegible gaps
D4 Repetition anomaly	A singular suffix should not double	M342 doubles in <1% of cases; consistent with reduplication patterns
D5 Punctuation attack	M342 is a scribal period mark	CV = 0.761; entropy delta = +0.703 bits on $N = 2,742$ (punctuation ruled out)
D6 Local guild symbols	Sign patterns are site-specific, not linguistic	Cross-site $Z = 50.12$, $p = 0$. All sites follow MD grammar. Geographic gradient confirms systemic transmission, not local symbology.
D7 Validation mark	M342 may be an administrative authority stamp	CV = 0.761 distinguishes suffix from stamp; Cross-Site Consistency ($Z = 50$) is inconsistent with a stamp applied independently at each site
D8 Commodity ledger	M342 is a standardised commodity tag	Commodity tags attach uniformly to all goods (low CV). M342 exhibits CV = 0.761, indicating selective bonding with specific signs. Furthermore, a commodity-ledger model does not explain the existence of strictly medial-locked signs (e.g., M99 at 100% medial for $L \geq 5$), which have no analogous function in inventory tracking.

Table 23: Known limitations with current status

Limitation	Status
Erosion bias (survivorship)	703 inscriptions dropped; clean corpus skews toward longer texts. Terminal rates may be slightly inflated for signs that occur in longer, well-preserved inscriptions.
RTL assumption	RESOLVED (Section 9). Direction codes from <code>indusarrays</code> Firestore assign 2,360 inscriptions as RTL (86.1%) and 175 as LTR (6.4%). LTR perplexity is $7,887\times$ below random under RTL grammar ($p = 1.03 \times 10^{-54}$), bounding the noise introduced by LTR inscriptions in Analyses 1–6.
West Asian sample size	13 inscriptions. Geographic gradient finding is robust at all other sites; West Asian score (+1.08) may partially reflect sampling noise.
No bilingual anchor	Phonetic readings probabilistic; none confirmed from bilingual text.
Unexplained M342 on seals	medial 23.9% of M342 occurrences on seals are medial (172 of 720 tokens). The multi-line text hypothesis does not fully account for these. Possible explanations include undetected physical line-breaks on damaged seals, compound morpheme boundaries, or a dual medial/terminal function for M342 in specific syntactic contexts. Decomposing this subset by seal carving style and per-face artifact records is deferred to future work.
Temporal gap (IVC → Keeladi: 1,300 yr)	Acknowledged; BRW tradition as bridge; transition probability verification pending machine-readable Megalithic data.
M342 vs M137 in literature	M342 is the primary jar sign; M137 is a variant form (jar with base), a distinct concordance entry. Secondary literature conflating the two should use M342 for the high-frequency terminal marker.
Sign list granularity (Allography)	The 398-sign inventory derives from Mahadevan (1977), which separates signs based on minor visual variants. If some signs are allographs (scribe-specific variants of the same sign), the true sign inventory is smaller. This would increase bigram frequencies and likely strengthen the observed transition constraints, but may alter the exact boundary thresholds of the functional classes in Analysis 5.
Segmentation error	The machine-readable corpus relies on Mahadevan’s manual segmentation of crowded seal impressions. Undetected segmentation errors (e.g., splitting one sign into two, or merging two adjacent signs) introduce noise into the bigram transition matrix.

17 Reproducibility

All code is available at <https://github.com/gkarun876/IVS> (MIT License).

Listing 1: Quick start

```
# mahadevan_corpus.json is pre-parsed and included in the repo

# Run all 9 analyses
python run_proofs_v2.py           # Analyses 1--4 (results/v2/)
python grammar_engine.py         # Analysis 5 (grammar_summary.txt)
python validation_mark_test.py   # Analysis 6 (cross_site_*.txt)
python direction_grammar_test.py # Analysis 7 (results/v4/)
python motif_correlation.py      # Analysis 8 (results/v4/)
python synonym_extractor.py      # Analysis 9 (results/v4/)

# Legacy pilot-corpus scripts
python indus_decode.py           # 41-seal pilot proofs
python typology_test.py          # Agglutinative vs Sanskrit
python adversarial_defense.py    # Defenses D1-D7
```

18 Conclusion

We have presented nine statistically independent structural tests on 2,742 inscriptions from the Indus Valley Civilisation.

1. **M342 is positionally constrained as a terminal morpheme** ($p = 1.70 \times 10^{-222}$), independently confirmed by ICIT’s TMK label.
2. **Positional entropy drops 0.558 bits** from initial to second slot, confirming an open initial vocabulary with terminal grammatical constraint.
3. **Signs M267, M99, M342 cluster at $5.5\times$ above chance** ($\chi^2 = 523$, $p \approx 0$), ruling out random co-occurrence.
4. **Sign M48 attaches to M342 at $8.97\times$ above independence**, consistent with selective suffix attachment.
5. **398 signs partition into three functional classes** (terminal-biased, initial-biased, medial-locked) with no cross-boundary signs above noise, consistent with morphological role differentiation.

6. **The Mohenjo-daro grammar generalises to all sites** ($Z = 50.12$, $p = 0$), with fit generally decaying with geographic distance—a pattern consistent with a standardized institutional writing system, not local iconography.
7. **The RTL assumption is resolved**: direction data from `indusarrays` confirms 86.1% RTL. LTR inscriptions score $7,887\times$ below random under RTL grammar ($p = 1.03 \times 10^{-54}$), bounding directional noise in Analyses 1–6.
8. **Sign M113 is $37.9\times$ positionally enriched on elephant-motif seals** and occupies the MEDIAL slot in a recurrent formula across Harappa and Lothal, structurally distinguishing it from terminal and initial signs; no semantic category is assigned. Sign pair M172–M304 appears in an identical eight-sign sequence on all 7 attested Kalibangan UNICORN seals (PMI = 7.94, zero sequence variation; $n = 7$, single site), consistent with a high-constraint institutional formula; no semantic or titular content is inferred.
9. **154 structural substitution synonyms** are recovered. M328 (90.1% Harappa-concentrated), M93 (100% Chanhudaro-concentrated), and M172 (87.5% Kalibangan-concentrated) are terminal-biased signs with strong single-site concentration. M328, M12, and M343 are confirmed positional equivalents (cross-site slot substitutions in identical formulae). Signs M304 and M328 have exact shape-matches in Tamil Nadu Iron Age graffiti [Rajan and Sivanantham, 2025], confirming the visual distinctiveness of these computationally isolated sign forms; no morphological or functional continuity is implied.

The hypothesis that the Indus Valley Script encodes random marks, administrative tokens, or site-specific guild symbology is statistically untenable given these results. The structural properties observed—positional constraint, selective suffix attachment, medial-positional bias, and cross-site grammatical standardisation—are consistent with a language encoding system and inconsistent with random or purely iconic marking; however, distinguishing a structured proto-writing system from a full language encoding system requires phonetic evidence beyond the scope of this study.

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